

Linus Torvalds Announces First Linux Kernel 6.5 Release Candidate

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Linus Torvalds announced today the general availability for public testing of the first Release Candidate (RC) development milestone of the upcoming Linux 6.5 kernel series. The two-week merge window, which opened with the release of Linux kernel 6.4 on June 25th, 2023, is now officially closed and the first Release Candidate is available for download for those of you who want to get an early taste of the next major release, Linux kernel 6.5. As expected, there will be many new features and improvements in Linux kernel 6.5. The final release of Linux kernel 6.5 is expected at the end of August or early September 2023, which depends on how many Release Candidate (RC) milestones Linus Torvalds will announce until then. Until then, if you want to take Linux kernel 6.5 for a test drive, you can download the first Release Candidate (RC1) milestone right now from either Linus Torvalds's git tree or from the kernel.org website.

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As expected, there will be many new features and improvements in Linux kernel 6.5. Some highlights include MIDI 2.0 support in ALSA, ACPI support for RISC-V, Landlock support for UML, better support for AMD "Zen" systems, Intel TPMI ((Topology Aware Register and PM Capsule Interface) support to the power capping subsystem and a TPMI interface driver for Intel RAPL, as well as user-space support for the ARMv8.8 memcpy/memset instructions.

Linux kernel 6.5 also promises to improve support for Lenovo Yoga Book yb1-x90f/1 and Nextbook Ares 8A tablets, add ACPI backlight quirks for Dell Studio 1569, Lenovo ThinkPad X131e (3371 AMD version), and Apple iMac11,3

laptops, improve performance for the Btrfs file system, and introduce SGX/HPE Ultraviolet support for Sub-NUMA clustering (SNC).

Of course, there will also be numerous new drivers for various devices to provide users with top-notch hardware support. These include the AHT20 temperature and humidity sensor, HP WMI sensors, and ROG Crosshair X670E Hero motherboard.

On top of that, Linux kernel 6.5 improves SMP scheduling's load balancer to recognize SMT cores with more than one busy sibling and allows lower-priority CPUs to pull tasks to avoid superfluous migrations, implements the "runnable boosting" feature in the EAS balancer to improve CPU utilization for certain workloads, and improves journalling, block allocator subsystems, and performance for parallel DIO overwrites on the EXT4 file system.

"Let's calm this party down. I have this slight suspicion that this may be one of those releases that may drag out, not because there are any particular issues I expect problems with, but simply due to a lot of Europe going on vacation for the month of August. But who knows... We'll see how things develop," said Linus Torvalds.

The final release of Linux kernel 6.5 is expected at the end of August or early September 2023, which depends on how many Release Candidate (RC) milestones Linus Torvalds will announce until then. It will be out on August 27th if there will be only seven RCs or on September 3rd if eight RCs are announced.

Until then, if you want to take Linux kernel 6.5 for a test drive, you can download the first Release Candidate (RC1) milestone right now from either Linus Torvalds's git tree or from the kernel.org website. However, I have to warn you not to install or use this development version on a production machine!

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